

46 YEARS OF EXCELLENCE

NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: SR.IIT _*CO-SC(MODEL-A&APEX) Date: 15-03-26
Time: 3 Hrs Max. Marks: 300

15-03-26_SR.IIT_STAR CO-SC(MODEL-A&APEX)_JEE MAIN_GTM-6_SYLLABUS

PHYSICS: TOTAL SYLLABUS

CHEMISTRY: TOTAL SYLLABUS

MATHEMATICS: TOTAL SYLLABUS

Rankers Academy JEE
THE NARAYANA GROUP

For More Join: @IITJEE_Advanced

PHYSICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. When an alternating voltage of 220 V is applied across a device P, a current of 0.25 A flows through the circuit and it leads the applied voltage by an angle $\frac{\pi}{2}$ radian. When the same voltage source is connected across another device Q, the same current is observed in the circuit but in phase with the applied voltage. What is the current when the same source is connected across a series combination of P and Q?

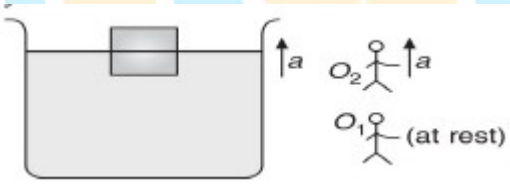
   A) $\frac{1}{4\sqrt{2}}$ A lagging in phase by $\frac{\pi}{4}$ with voltage

   B) $\frac{1}{4\sqrt{2}}$ A leading in phase by $\frac{\pi}{4}$ with voltage

   C) $\frac{1}{\sqrt{2}}$ A leading in phase by $\frac{\pi}{4}$ with voltage

   D) $\frac{1}{4\sqrt{2}}$ A leading in phase by $\frac{\pi}{2}$ with voltage

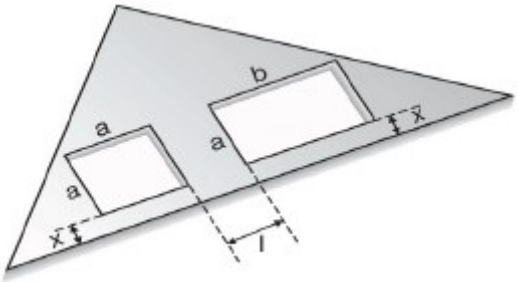
2. A block is partially immersed in a liquid and the vessel is accelerating upwards with an acceleration a. The block is observed by two observers O_1 and O_2, one at rest and the other accelerating with an acceleration a upward. The buoyant force on the block is



   A) Same for O_1 and O_2 B) greater for O_1 than O_2

   C) Greater for O_2 than O_1 D) data is not sufficient

3. A triangular plate has two cavities, one square and one rectangular as shown. The plate is heated



   A) a increases, b decreases B) a and b only increase

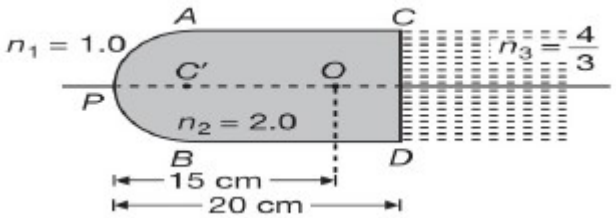
   C) a and b increase, x and l decrease D) a, b, x and l all increase

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4. A tuning fork, whose frequency as given by the manufacturer 512 Hz, is being tested using an accurate oscillator. It is found that they produce 2 beasts per second when the oscillator reads 514 Hz, and 6 beats per second when it reads of 510 Hz. The actual frequency of the fork is

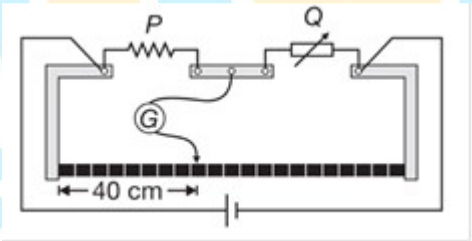
A) 508 Hz B) 512 Hz C) 516 Hz D) 518 Hz

5. The slab of a material of refractive index 2 shown in figure has a curved surface APB of radius of curvature 10 cm and a plane surface CD. On the left of APB is air and on the right of CD is water with refractive indices as given in figure. An object O is placed at a distance of 15 cm from pole P as shown. The distance of the final image of O from P, as viewed from the left is [R.I of air =1, R.I of glass =2,R.I of liquid $=\frac{4}{3}$]



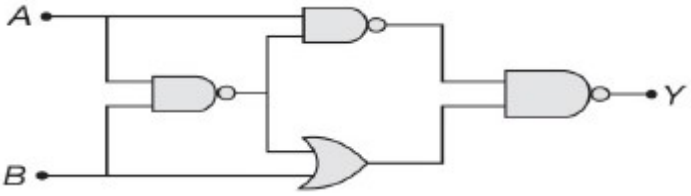
A) 20 cm B) 30 cm C) 40 cm D) 50 cm

6. In a meter bridge, the gaps are closed by two resistances P and Q and the balance point is obtained at 40 cm. When Q is shunted by a resistance of 10Ω, the balance point shifts to 50 cm. The values of P and Q are



A) $\frac{10}{3}\Omega, 5\Omega$ B) $20\Omega, 30\Omega$ C) $10\Omega, 15\Omega$ D) $5\Omega, \frac{15}{2}\Omega$

7. The output of the given logic circuit is



A) $A\bar{B}+\bar{A}B$ B) $A\bar{B}$ C) $AB+\overline{AB}$ D) $\bar{A}B$

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8. **Statement – 1:** A heavy weight is suspended from the spring. A person rises the weight slowly till the spring acquire its natural length. The work done by person is W. The energy stored in the stretched spring was E. The gain in gravitational potential energy is $(W+E)$

Statement – 2: Work done by the spring force is always negative

A) If both statements are TRUE and **Statement – 2** is the correct explanation of **Statement - 1**

B) If both statements are TRUE and **Statement – 2** is not the correct explanation of **Statement – 1**

C) If **Statement – 1** is TRUE and **Statement – 2** is FALSE

D) If **Statement – 1** is FALSE but **Statement – 2** is TRUE

9. A, B, C and D are four different physical quantities having different dimensions. None of them is dimensionless. But we know that the equation $AD = C\ln(BD)$ holds true. Then which of the combination is not a meaningful quantity?

A) $\frac{C}{BD}-\frac{(AD)^2}{C}$ B) $A^2-B^2C^2$ C) $\frac{A}{B}-C$ D) $\frac{(A-C)}{D}$

10. Match the quantities in **COLUMN – I** with their dependence on the principal quantum number n and the atomic number Z in an orbit in **COLUMN – II** for hydrogen or hydrogen like ions.

COLUMN – I		**COLUMN – II**	
A)	Angular speed of electron	p)	$\frac{n^3}{Z^2}$
B)	Time period of electron	q)	n
C)	Angular momentum of electron	r)	$\frac{Z^2}{n^3}$
D)	Magnetic moment of electron	s)	$\frac{Z^3}{n^5}$
E)	Magnetic field at the nucleus		

A) $A-p;B-r;C-q;D-p;E-s$ B) $A-r;B-p;C-q;D-s;E-r$

C) $A-s;B-p;C-q;D-q;E-r$ D) $A-r;B-p;C-q;D-q;E-s$

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11. **Statement – 1:** The bob of a simple pendulum is a hollow ball full of water. If a fine hole is made at the bottom of the ball, then the time period will vary till the ball gets empty and then becomes constant.

Statement – 2: The time period of simple pendulum does not depend upon mass

A) If both statements are TRUE and **Statement – 2** is the correct explanation of **Statement - 1**

B) If both statements are TRUE but **Statement – 2** is not the correct explanation of **Statement – 1**

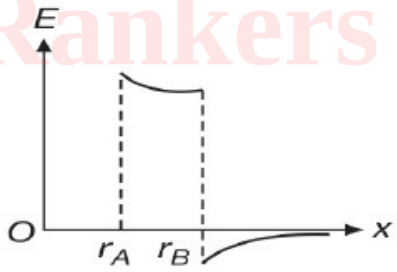
C) If **Statement – 1** is TRUE and **Statement – 2** if FALSE

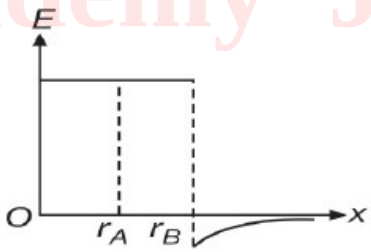
D) If **Statement – 1** is FALSE but **Statement – 2** is TRUE

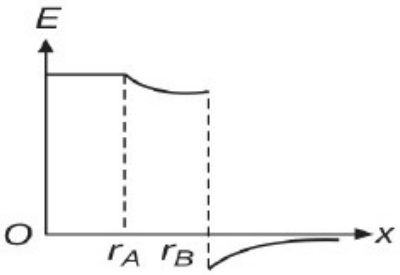
12. A black body radiates power P and maximum energy is radiated by it around a wavelength λ_0. The temperature of the black body is now changed such that it radiates maximum energy around the wavelength $\frac{3\lambda_0}{4}$. The power radiated by it now is

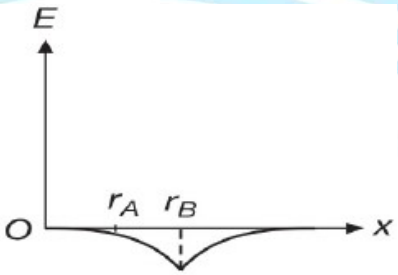
A) $\frac{256}{81}P$ B) $\frac{16}{9}P$ C) $\frac{64}{27}P$ D) $\frac{4}{3}P$

13. Two concentric conducting thin spherical shells A, and B having radii r_A and $r_B\,(r_B > r_A)$ are charged to Q_A and $-Q_B\,(|Q_B| > |Q_A|)$. The variation of electric filed (E) with distance r from centre is best presented by

A) 

B) 

C) 

D) 

14. Sun radiates energy in all direction. The average energy received at earth is $1.4\,kWm^{-2}$. The average distance between the earth and the sun is $1.5\times10^{11}\,m$. If this energy is released by conversion of mass into energy, then the mass lost per day by the sun is approximately

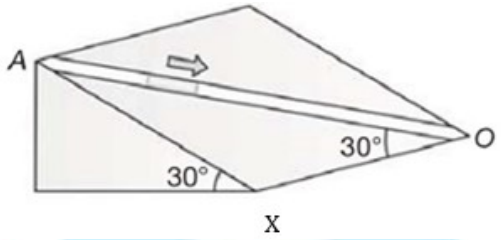
A) $4.4\times10^{9}\,kg$ B) $7.6\times10^{14}\,kg$ C) $3.8\times10^{12}\,kg$ D) $3.8\times10^{14}\,kg$

For More Join: @IITJEE_Advanced

15. Light with an energy flux of $20\,W/cm^2$ falls on a non-reflecting surface at normal incidence. If the surface has an area of $30\,cm^2$. The total momentum delivered (for complete absorption) during 30 minutes is

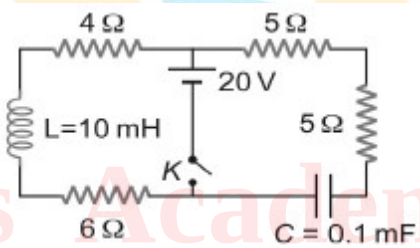
A) $36\times10^{-5}\,kg-m/s$ B) $36\times10^{-4}\,kg-m/s$ C) $1.08\times10^{4}\,kg-m/s$ D) $1.08\times10^{7}\,kg-m/s$

16. An inclined plane makes an angle 30° with the horizontal. A groove OA = 5 m cut in the plane makes an angle 30° with OX. A short smooth body is free to slide down the influence of gravity. The time taken by the body to reach from A to O is $\left(g=10\,ms^{-2}\right)$



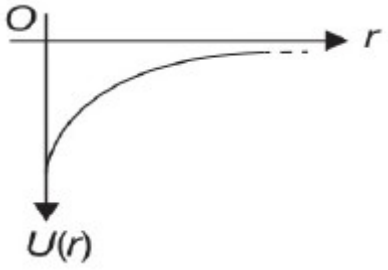
A) 4s B) 2s C) $2\sqrt{2}s$ D) 1s

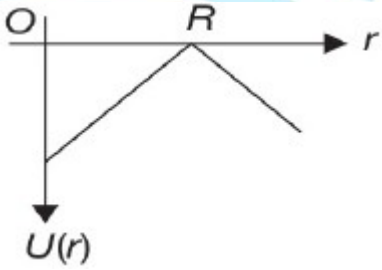
17. In the circuit shown, the key (K) is closed at $t=0$, the current through the key at the instant $t=10^{-3}\ln 2$, is

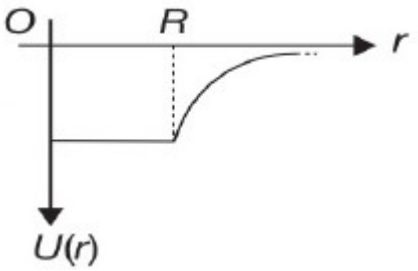


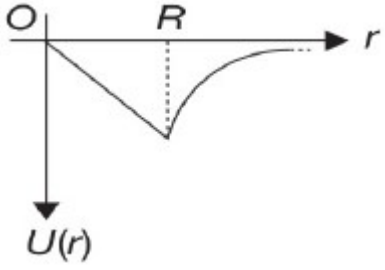
A) 2 A B) 8 A C) 4 A D) Zero

18. A shell of mass M and radius R has a point mass m placed at a distance r from its centre. The gravitational potential energy $U(r)$ vs r will be

A)


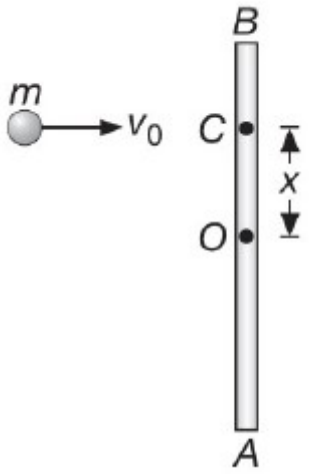
B)


C)


D)


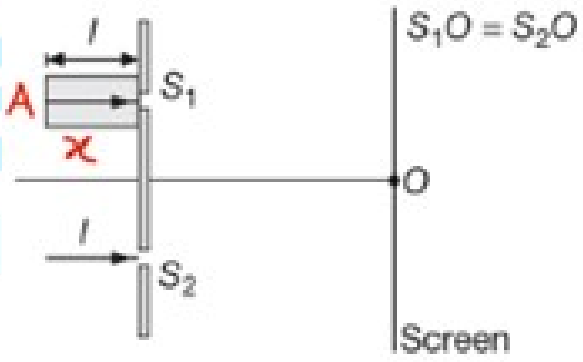
For More Join: @IITJEE_Advanced

19. A uniform rod AB of length L having mass m are lying on a smooth table. A small particle of mass m strikes the rod with a velocity υ_0 at point C, a distance x from the centre O. The particle comes to rest after collision. The value of x, so that point A of the rod remains stationary just after collision is



A) $\frac{L}{3}$ B) $\frac{L}{6}$ C) $\frac{L}{4}$ D) $\frac{L}{12}$

20. In the figure shown, a parallel beam of light is incident on the plane of the slits of a Young's double slit experiment. Light incident on the slit, S_1 passes through a medium of variable refractive index $\mu = 1 + ax$ (where x is the distance from the point A as shown), up to a distance ℓ before falling on S_1. The entire remaining space is filled with air. If at O, a minima is formed, then the minimum value of the positive constant a (in terms of ℓ and wavelength λ in air) is



A) $\frac{\lambda}{\ell}$ B) $\frac{\lambda}{\ell^2}$ C) $\frac{\ell^2}{\lambda}$ D) $\frac{\lambda}{2\ell^2}$

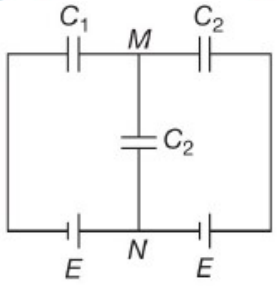
For More Join: @IITJEE_Advanced

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme:
Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

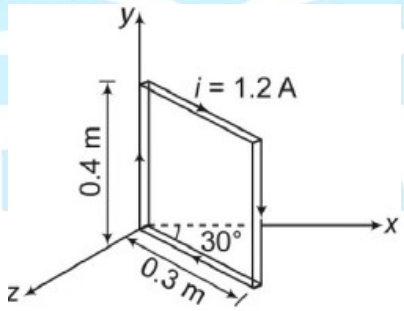
21. A golfer standing on level ground hits a ball with a velocity of $u = 52 ms^{-1}$ at an angle θ above the horizontal. If $\tan\theta = \frac{5}{12}$, then the time for which the ball is at least 15 m above the ground will be (in seconds) (take $g = 10 ms^{-2}$)

22. Find the potential difference, in volt, between points M and N of the system shown, if the emf is equal to $E = 110 V$ and the capacitance ratio $\frac{C_2}{C_1}$ is 2



23. The two perpendicular faces of a right isosceles prism are coated with silver. The rays incident at an arbitrary angle on the hypotenuse face emerge from the prism after suffering a deviation of x degree. Find x.

24. A rectangular loop consists of $N = 100$ closed wrapped turns and has dimensions $(0.4 m \times 0.3 m)$. The loop is hinged along the y-axis and its plane makes an angle $\theta = 30^\circ$ with the x-axis. The magnitude of the torque exerted on the loop by a uniform magnetic field B = 0.8 T directed along the x-axis when current is $i = 1.2 A$ in the direction shown is $(2.4)^p (3)^q N-m$. Find pq.



25. Two balls A and B having different but unknown masses, collide elastically. A is initially at rest when B has a speed υ. After collision B has a speed $\frac{\upsilon}{2}$ and moves at right angles to its original motion. The direction in which ball A moves after collision with respect to initial direction of motion of B is $\tan^{-1}\left(\frac{1}{\alpha}\right)$. Calculate α.

For More Join: @IITJEE_Advanced

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

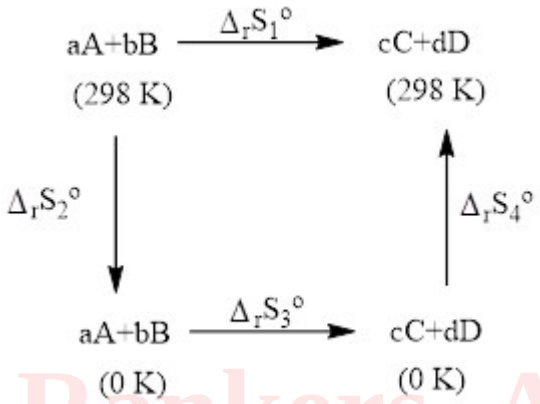
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

26. H_3PO_4 undergoes dissociation as $H_3PO_4 \longrightarrow 3H^{\oplus} + PO_4^{3-}$. If $\frac{n_{cation} + n_{anion}}{n_{H_3PO_4}(\text{undissociated})} = \frac{8}{3}$ then, percentage dissociation of H_3PO_4 will be (where n_{cation} = number of cation)

A) 40% B) 20% C) 80% D) 60%

27. Which of the following is incorrect for the change shown below?

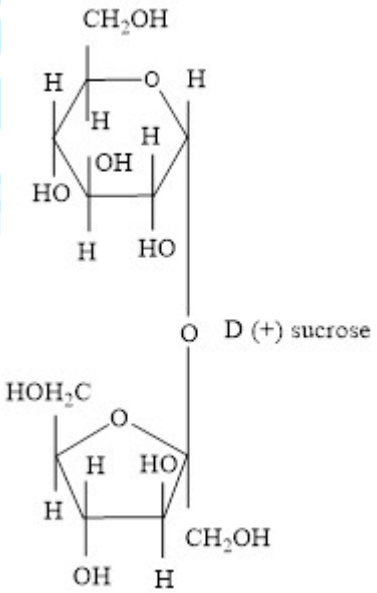
[A, B, C and D have perfect crystal structures]



A) $\Delta_r S_1^\circ = \left[cS_c^\circ + dS_D^\circ\right] - \left[aS_A^\circ + bS_B^\circ\right]$ B) $\Delta_r S_2^\circ = -\left[aS_A^\circ + bS_B^\circ\right]_{298K}$

C) $\Delta_r S_3^\circ = 0$ D) $\Delta_r S_4^\circ = -\left[cS_c^\circ + dS_D^\circ\right]_{298K}$

28. The number of chiral centres present in the following compound



A) 7 B) 8 C) 9 D) 10

For More Join: @IITJEE_Advanced

29. In a sample of hydrogen atoms all electrons are in a particular excited state n, when all electrons returned to ground state, photons with 6 different wavelengths are emitted. Which of the following is correct?

A) Out of 6 different photons only 2 photons have speed equal to that of visible light

B) If highest energy photon emitted from above sample is incident on the metal plate having work function 8 eV, K.E. of liberated photo-electron may be equal to or more than 4.75 eV

C) Total number of radial nodes in 'ns' orbital is 3

D) Total number of radial nodes in (n – 1)p orbital is 2

30. Consider the exothermic reaction

$$[CoCl_4]^{2-}(aq) + 6H_2O(\ell) \rightleftharpoons [Co(H_2O)_6]^{2+}(aq) + 4Cl^-(aq)$$

[Consider initially only $[CoCl_4]^{2-}$ is added to water]

Which of the following is incorrect

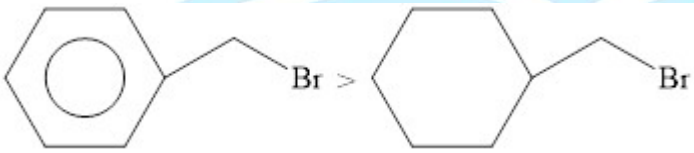
A) If HCl is added to above equilibrium then concentration of $[CoCl_4]^{2-}(aq)$ will increase

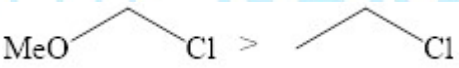
B) On dilution with water the concentration of $[CoCl_4]^{2-}(aq)$ will decrease

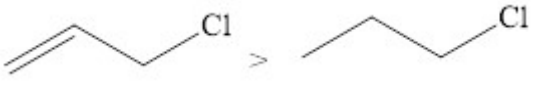
C) On doubling the volume of solution by addition water the final concentration of $[Co(H_2O)_6]^{2+}(aq)$ will be more than its initial concentration

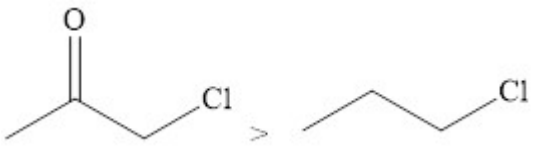
D) On increasing temperature the concentration of $[CoCl_4]^{2-}$ will increase

31. In the following the rate of both S_N1 and S_N2 follows the same order except in_

A) 

B) 

C) 

D) 

For More Join: @IITJEE_Advanced

32. The standard reduction potential data at $25^{\circ}C$ is given below:

$E^{\circ}\left(Fe^{3+}, Fe^{2+}\right) = +0.77V; E^{\circ}\left(Fe^{2+}, Fe\right) = -0.44V$

$E^{\circ}\left(Cu^{2+}, Cu\right) = +0.34V; E^{\circ}\left(Cu^{+}, Cu\right) = +0.52V$

$E^{\circ}\left[O_2(g) + 4H^{+} + 4e^{-} \longrightarrow 2H_2O\right] = +1.23V;$

$E^{\circ}\left[O_2(g) + 2H_2O + 4e^{-} \longrightarrow 4OH^{-}\right] = +0.40V$

$E^{\circ}\left(Cr^{3+}, Cr\right) = -0.74V; \quad E^{\circ}\left(Cr^{2+}, Cr\right) = -0.91V$

Match E° of the redox pair in **Column – I** with the values given in **Column – II**

Column – I		Column – II	
A)	$E^{\circ}\left(Fe^{3+}, Fe\right)$	1)	- 0.18 V
B)	$E^{\circ}\left(4H_2O \rightarrow 4H^{+} + 4OH^{-}\right)$	2)	- 0.4 V
C)	$E^{\circ}\left(Cu^{2+} + Cu \longrightarrow 2Cu^{+}\right)$	3)	- 0.04 V
D)	$E^{\circ}\left(Cr^{2+}, Cr^{2+}\right)$	4)	- 0.83 V

A) $A \rightarrow 2; B \rightarrow 3; C \rightarrow 1; D \rightarrow 4$ B) $A \rightarrow 1; B \rightarrow 2; C \rightarrow 3; D \rightarrow 4$

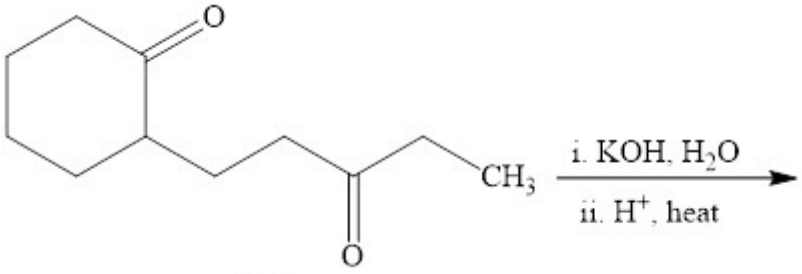
C) $A \rightarrow 3; B \rightarrow 4; C \rightarrow 1; D \rightarrow 2$ D) $A \rightarrow 4; B \rightarrow 3; C \rightarrow 1; D \rightarrow 2$

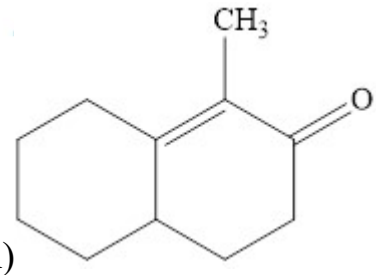
33. When Br_2 is treated with aqueous solutions of $NaF, NaCl$ and NaI separately

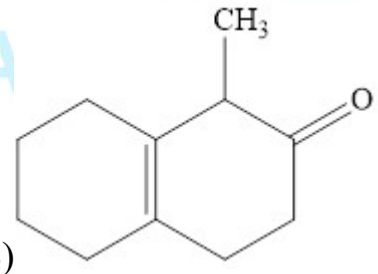
A) F_2, Cl_2 and I_2 are liberated B) only F_2 and Cl_2 are liberated

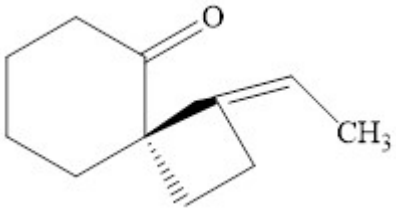
C) only I_2 is liberated D) only Cl_2 is liberated

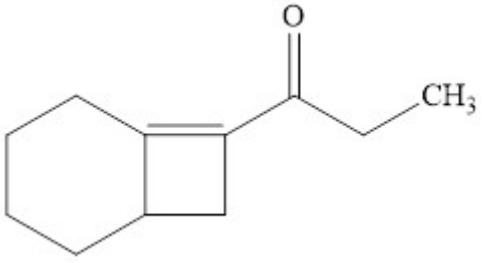
34. The major product of the following reaction is



A) 

B) 

C) 

D) 

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35. **Statement – 1:** Treatment of D-glucose with dilute alkali affords an equilibrium mixture consisting of D-mannose, D-fructose and starting substance D-glucose.

Statement – 2: The reaction involves an intermediate in which hybridization of C_2 changes from

sp^3 and sp^2

A) Statement – 1 is correct and Statement – 2 is incorrect

B) Statement – 1 is incorrect and Statement – 2 is correct

C) Both Statements – 1 and Statement – 2 are correct

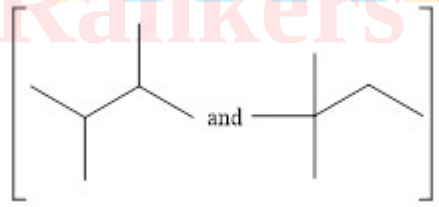
D) Both Statements – 1 and Statement – 2 are wrong

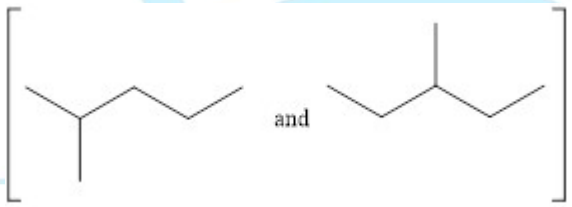
36. $\left[NiCl_2\left\{P(C_2H_5)_2(C_6H_5)\right\}_2\right]$ exhibits temperature dependent magnetic behaviour (paramagnetic/ diamagnetic). The coordination geometries of Ni^{2+} in the paramagnetic and diamagnetic states are respectively

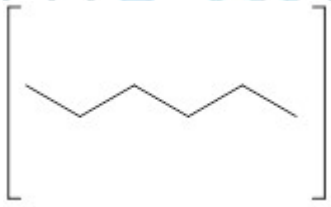
A) tetrahedral and tetrahedral B) square planar and square planar

C) tetrahedral and square planar D) square planar and tetrahedral

37. Isomers of hexane, based on their on their branching, can be divided into three distinct classes as shown in the figure

I. 

II. 

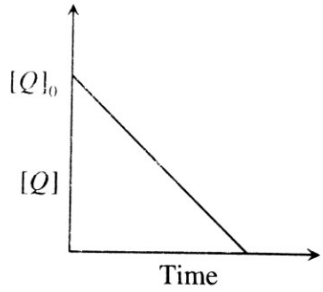
III. 

The correct order of their boiling point is

A) I > II > III B) III > II > I C) II > III > I D) III > I > II

For More Join: @IITJEE_Advanced

38. In the reaction, $P + Q \longrightarrow R + S$ the time taken for 75% reaction of P is twice the time taken for 50% reaction of P. The concentration of Q varies with reaction time as shown in the figure. The overall order of the reaction is



A) 2 B) 3 C) 0 D) 1

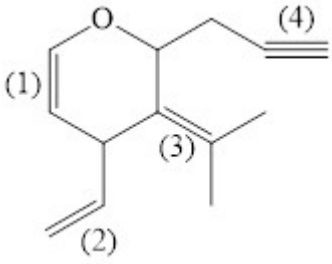
39. Match the compounds/ions in **Column – I** with their properties/reactions in **Column – II**

	Column – I		**Column – II**
A)	C_6H_5CHO	1)	gives precipitate with 2,4-dinitrophenyl-hydrazine
B)	$CH_3C \equiv CH$	2)	gives precipitate with ammonical $AgNO_3$
C)	CN^-	3)	is a nucleophile
D)	I^-	4)	is involved in cyanohydrin formation

A) $A \rightarrow 1,4; B \rightarrow 2; C \rightarrow 2,3,4; D \rightarrow 2,3$ B) $A \rightarrow 1,3,4; B \rightarrow 2,3; C \rightarrow 1,2,3; D \rightarrow 2$

C) $A \rightarrow 1,3; B \rightarrow 1,4; C \rightarrow 1,3; D \rightarrow 2,3$ D) $A \rightarrow 2,3; B \rightarrow 1,2; C \rightarrow 1,3; D \rightarrow 1,2$

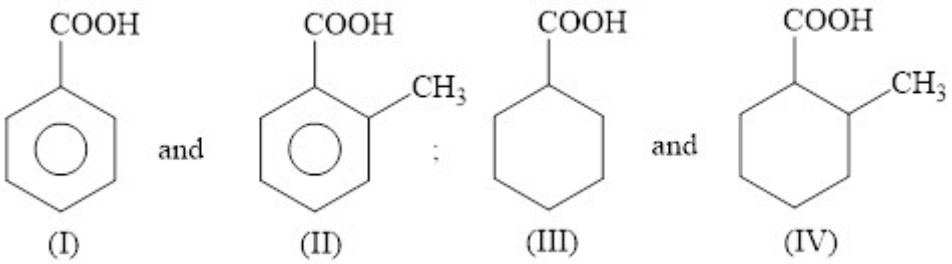
40. The correct reactivity order of following $C = C / C \equiv C$ bonds towards $Br^{\oplus}$ is



A) 4 > 3 > 2 > 1 B) 3 > 2 > 1 > 4 C) 1 > 3 > 4 > 2 D) 1 > 3 > 2 > 4

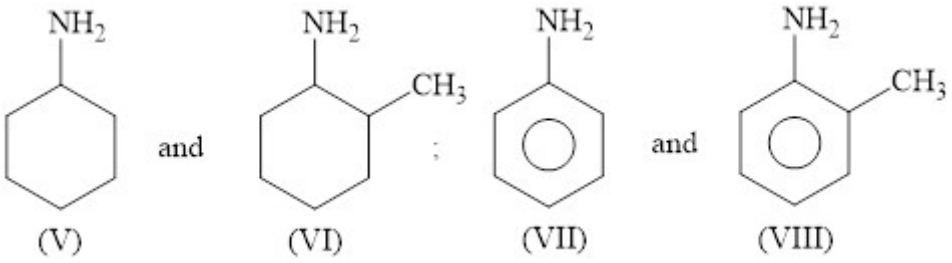
41. The correct order of acid and basic strength for the following pair of compounds should be:

Acid strength:



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Basic strength:



(V) (VI) (VII) (VIII)

A) I > II; III > IV; V < VI; VII < VIII

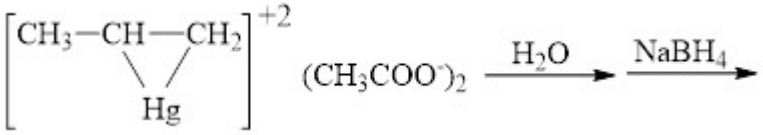
B) I < II; III > IV; V < VI; VII > VIII

C) I > II; III > IV; V > VI; VII > VIII

D) I < II; III > IV; V < VI; VII < VIII

42. In which of the following reactions alcohol will not be formed as the final product?

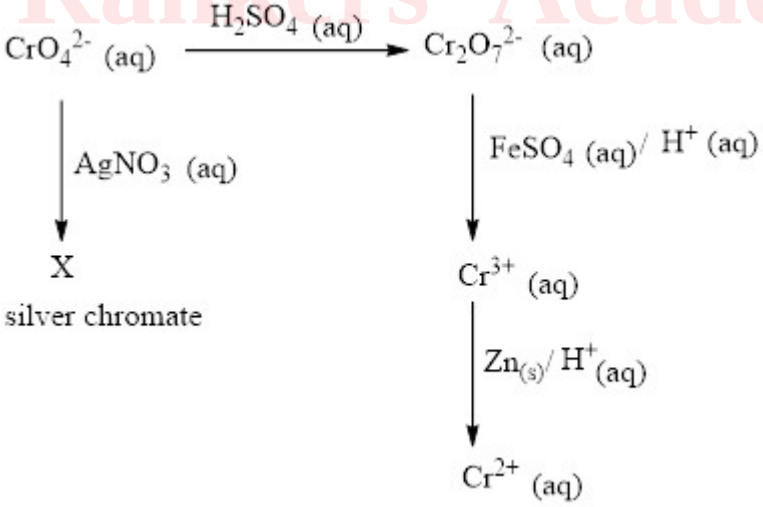
A) $(CH_3CH_2CH_2-)_3B \xrightarrow{H_2O_2+NaOH}$

B) 

C) $CH_3-CH=CHMgBr \xrightarrow{O_2,\Delta} \xrightarrow{H_2O}$

D) $CH_3-CH=CH_2MgBr + \triangledown O \xrightarrow{THF} \xrightarrow{H_3O^+}$

43. Consider the reactions shown below.



Which of the following statements is false? [Atomic mass of zinc = 65.4]

A) silver chromate has the formula Ag_2CrO_4

B) the minimum mass of zinc required to reduce 0.100 mole of Cr^{3+} to Cr^{2+} is 6.54 g

C) the conversion of CrO_4^{2-} into $Cr_2O_7^{2-}$ is not a redox reaction

D) the equation, $Cr_2O_7^{2-} + 14H^+ + 6Fe^{2+} \rightarrow 6Fe^{3+} + 2Cr^{3+} + 7H_2O$

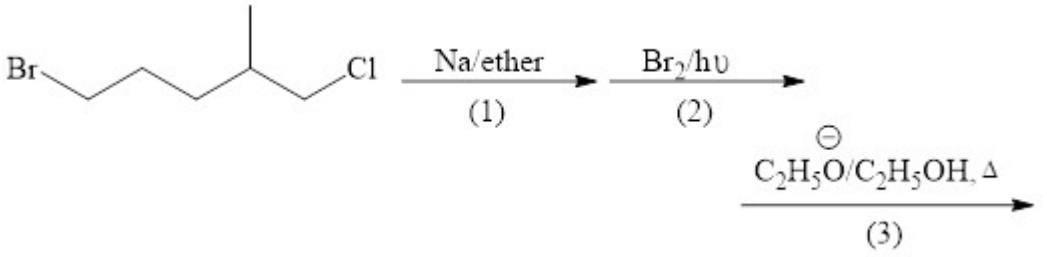
correctly describes the reduction of $Cr_2O_7^{2-}$ by acidified $FeSO_4$

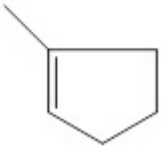
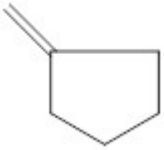
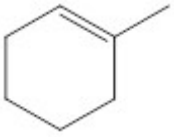
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44. In which of the following compounds B atoms are in sp^2 and sp^3 hybridisation states?

A) borax B) diborane C) borazole D) all of these

45. The major product of the following reaction is



A)  B)  C)  D) 

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only.** **Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme:
Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

46. Total number of species having bond order greater than that of nitric oxide molecule is/are

$CO, N_2, N_2^-, C_2^-, F_2^+, S_2, O_2^-$

47. In the oxidation of Phosphorus with excess Oxygen 0.2 mole of P_4 produces how many grams of P_4O_{10} (Round off to nearest whole number)

48. Consider the following reaction:

$$Na_3PO_4 + (NH_4)_2\,MoO_4 + HNO_3\,(dil) \rightarrow \underset{\text{"canary yellow ppt."}}{'P'}$$

Total number of O atoms in the molecular formula of P is/are _____.

49. Among the triatomic molecules/ions, $BeCl_2, N_3^-, N_2O, NO_2^+, SCl_2, ICl_2^-, I_3^-$ and XeF_2, the total number of linear molecule(s)/ion(s) where the hybridization of the central atom does not have contribution from the d-orbital(s) is

[Atomic number: S = 16, Cl = 17, I = 53 and Xe = 54]

50. A decapeptide (Mol. Wt. 796) on complete hydrolysis gives glycine (Mol. Wt. 75), alanine and phenylalanine. Glycine contributes 47.0% to the total weight of the hydrolysed products. The number of glycine units present in the decapeptide is

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MATHEMATICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

51. Let $I_1 = \int_0^{\pi} \frac{x \sin x \cos^4 x}{\left(1+\sin^2 x\right)^3} dx$, $I_2 = \int_0^1 \frac{x^4}{\left(2-x^2\right)^3} dx$ Then $\frac{I_1}{I_2} =$

A) $\frac{\pi}{2}$ B) $\frac{2}{\pi}$ C) π D) $\frac{1}{\pi}$

52. Let the foci of hyperbola coincide with the foci of the ellipse $\frac{x^2}{36}+\frac{y^2}{16}=1$. If the eccentricity of the hyperbola is 56, then the length of its latus rectum is

A) 12 B) 16 C) $\frac{96}{\sqrt{5}}$ D) $24\sqrt{5}$

53. Let $y = y_1(x)$ and $y = y_2(x)$ be two distinct solutions of the differential equation $\frac{dy}{dx} = x + y$, with $y_1(0) = 0$ and $y_2(0) = 1$ respectively. Then, the number of points of intersection of $y = y_1(x)$ and $y = y_2(x)$ is

A) 0 B) 1 C) 2 D) 3

54. If the line $x = \alpha$ divides the area of region $R = \left\{(x, y) \in \mathbb{R}^2 : x^3 \le y \le x, 0 \le x \le 1\right\}$ into two equal parts, then

A) $0 < \alpha \le \frac{1}{2}$ B) $\frac{1}{2} < \alpha < 1$ C) $2\alpha^4 + 4\alpha^2 + 1 = 0$ D) $\alpha^4 + 4\alpha^2 - 1 = 0$

55. Let $L_1 : \frac{x-1}{2} = \frac{y-2}{3} = \frac{z+1}{4}$ and $L_2 : \frac{x+2}{1} = \frac{y-3}{2} = \frac{z-1}{3}$ be two skew lines. A line L_3 intersects L_1 at P and L_2 at Q such that $\overrightarrow{PQ}$ is the line of shortest distance between them. If $R(\alpha, \beta, \gamma)$ lies on L_3 such that $PR : RQ = 2 : 3$ internally, then $14|\alpha + \beta + \gamma|$ is:

A) 544 B) 602 C) 89 D) 103

For More Join: @IITJEE_Advanced

56. The sum of the absolute maximum and absolute minimum values of the function $f(x)=\frac{x^2+2x+1}{x^2-x+2}$ on the interval $[0,3]$ is:

A) $\frac{1}{2}$ B) $\frac{17}{7}$ C) $\frac{39}{14}$ D) $\frac{45}{14}$

57. The coordinates of the orthocenter of the triangle formed by the lines $x+y=1, 2x+3y=6$ and $4x-y+4=0$ are:

A) $(-3,7)$ B) $\left(\frac{3}{7},\frac{22}{7}\right)$ C) $\left(-\frac{3}{7},\frac{27}{7}\right)$ D) $(2,-5)$

58. Number of elements in the domain of function $f(x)=3\sin^{-1}x^2+\frac{(7x+1)!}{\sqrt{x+1}}$ is

A) 5 B) 6 C) 8 D) 9

59. If the locus of the centre of the variable circle which intersects the circles $x^2+y^2=4$ and $x^2+y^2+2x+4y=6$ orthogonally has the equation $ax+by=1$, then the value of $\int_a^b\left(\log_e\left|\sin\frac{\pi x}{2}\right|\right)dx$ is equal to

A) $\log_e 2$ B) $-\log_e 2$ C) $\frac{\pi}{2}\log_e 2$ D) $\frac{-\pi}{2}\log_e 2$

60. If $\lim_{x\to 0}\frac{\alpha e^x+\beta e^{-x}+\gamma\sin x}{x\sin^2 x}=\frac{2}{3}$, where $\alpha,\beta,\gamma\in R$, then which of the following is not correct?

A) $\alpha^2+\beta^2+\gamma^2=6$ B) $\alpha\beta+\beta\gamma+\gamma\alpha+1=0$

C) $\alpha\beta^2+\beta\gamma^2+\gamma\alpha^2+3=0$ D) $\alpha^2-\beta^2+\gamma^2=4$

61. Sum of the values of x for which $\{x+5\},[x],x-1$ are in AP (where $\{.\}$ denotes FPF and $[.]$ denotes GIF) is:

A) $\frac{1}{2}$ B) $-\frac{1}{2}$ C) 1 D) – 1

62. Let $\vec{a}=2\hat{i}-5\hat{j}-5\hat{k}$ and $\vec{b}=\hat{i}-\hat{j}+3\hat{k}$. If $\vec{c}$ is a vector such that $2(\vec{a}\times\vec{c})+3(\vec{b}\times\vec{c})=\vec{0}$ and $(\vec{a}-\vec{b}).\vec{c}=-67$, then $|\vec{c}\times\hat{k}|^2$ is equal to

A) 193 B) 233 C) 218 D) 205

For More Join: @IITJEE_Advanced

63. If $f(x)=x+1, g^{-1}(x)=x^3+x+1$ and $g(f(x))=h(x)$, then $\lim\limits_{x\to 1}\dfrac{h^{-1}(x)-h^{-1}(1)}{x-1}$ is equal to

A) 2 B) 4 C) 1 D) 8

64. Let function $f(x)$ defined from $\mathbb{R}-\{0\}\to\mathbb{R}$ satisfies $x^2f^1(x)+2xf(x)=e^x$ and $f(2)=\dfrac{e^2}{4}$.

Which of the following is INCORRECT

A) $f(x)=1$ has exactly one real solution

B) $f(x)=3$ has exactly three real solutions

C) $f(x)$ has local maxima but no local minima

D) $f(x)$ has local minima but no local maxima

65. Two lines L_1 and L_2 passing through the point $P(1,2)$ cut the line $x+y=4$ at a distance of $\dfrac{\sqrt{6}}{3}$ units from P. then, the angles made by L_1, L_2 with positive X-axis are

A) $\dfrac{\pi}{3},\dfrac{\pi}{6}$ B) $\dfrac{\pi}{8},\dfrac{3\pi}{8}$ C) $\dfrac{\pi}{12},\dfrac{5\pi}{12}$ D) $\dfrac{\pi}{4},\dfrac{\pi}{8}$

66. The chord AC of the parabola $y^2=4ax$ subtends an angle of 90^0 at points B and D on the parabola. If points A, B, C and D are represented by $(at_i^2, 2at_i), i=1,2,3,4$ respectively, then the value of $\left|\dfrac{t_2+t_4}{t_1+t_3}\right|$.

A) 1 B) 2 C) 3 D) 4

67. Let the system of equations

$x+5y-z=1$

$4x+3y-3z=7$

$24x+y+\lambda z=\mu$

$\lambda,\mu\in R$, have infinitely many solutions. Then the number of the solutions of this system, if x, y, z are integers and satisfy $7\le x+y+z\le 77$, is:

A) 3 B) 6 C) 5 D) 4

For More Join: @IITJEE_Advanced

68. Let $f(x)$ be a cubic polynomial on R which increases in the interval $(-\infty,0)$ and in $(1,\infty)$ and decreases in the interval $(0,1)$. If $f'(2)=6$ and $f(2)=2$, then the value of $\tan^{-1}(f(1))+\tan^{-1}\left(f\left(\frac{3}{2}\right)\right)+\tan^{-1}(f(0))$ is equal to:

A) $\tan^{-1}2$ B) $\cot^{-1}2$ C) $-\tan^{-1}2$ D) $-\cot^{-1}2$

69. Two Grand Masters Anand and Kasparov, play 3 games of chess. The probability that Anand wins a game is 0.5, looses with probability 0.3 and ties with probability 0.2. If he plays 3 games then chance that he wins at least two games, is:

A) 0.3 B) 0.4 C) 0.5 D) 0.6

70. Let z_1, z_2 and z_3 be complex numbers such that $|z_1|=|z_2|=|z_3|=|z_1+z_2+z_3|=3$. If $|z_1-z_2|=|z_1-z_3|$ and $z_2\neq z_3$ then the value of $|z_1+z_2|.|z_1+z_3|$ is

A) 9 B) 12 C) 36 D) 18

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only.** **Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme:
Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

71. Let $L=\lim_{x\to 0}\left(\frac{(1+x)^{1/x}}{e}\right)^{\frac{1}{x}}$. Find the value of $2|\ln(L)|$

72. A circle C passes through the focus S of the parabola $y^2=16x$ and touches the parabola at the point $P(4,8)$. The circle C intersects the positive x-axis at points S and Q. A focal chord of the parabola is drawn parallel to the tangent to the circle C at point Q. If the lengths of this focal chord is L, find the value of $L/4$.

73. Assuming all the terms of A.P $[a_n]$ are integers with $a_1=2019$ and for $n\in I^+$ there always exists positive integer m such that $a_1+a_2+a_3+...+a_{n-2}+a_{n-1}+a_n=a_m$, then the number of such sequences is __

74. If the area bounded by the curves $x^2=y, x^2=-y$ and $y^2=4x-3$ is $\frac{m}{n}$, where m and n are relatively prime positive integer, then $|m-n|$ is

75. Let $A=\{2,3,4,5,6,7,8\}$ and $B=\{3,6,7,9\}$. If the number of elements in the set $\{C\subseteq A: C\cap B\neq\phi \text{ and } 8\in C\}$ is N then the units digit of N is

For More Join: @IITJEE_Advanced